# Problem 2

Find the points corresponding to the complex numbers and determine their position in the complex plane:

$$1^\circ\ i;\quad 2^\circ\ -2i;\quad 3^\circ\ -1+i;\quad 4^\circ\ 2-3i$$

# Solution

We will convert each complex number $z = x + iy$ to its corresponding point $(x, y)$ and identify its location in the complex plane.

## Recall: Complex Plane Representation

A complex number $z = x + iy$ corresponds to the point $(x, y)$ where:

- $x = \text{Re}(z)$ is plotted on the horizontal (real) axis
- $y = \text{Im}(z)$ is plotted on the vertical (imaginary) axis

The complex plane is divided into four quadrants:

- **Quadrant I:** $x > 0, y > 0$ (positive real, positive imaginary)
- **Quadrant II:** $x < 0, y > 0$ (negative real, positive imaginary)
- **Quadrant III:** $x < 0, y < 0$ (negative real, negative imaginary)
- **Quadrant IV:** $x > 0, y < 0$ (positive real, negative imaginary)

## Part $1^\circ$: $z = i$

### Extract Real and Imaginary Parts

$$z = i = 0 + i \cdot 1$$

- Real part: $x = 0$
- Imaginary part: $y = 1$

### Corresponding Point

$$\boxed{(0, 1)}$$

### Position in Complex Plane

The point $(0, 1)$ lies on the **positive imaginary axis**.

Since $x = 0$ and $y > 0$, it is not in any quadrant but on the boundary between Quadrants I and II.

### Additional Properties

- Modulus: $|i| = \sqrt{0^2 + 1^2} = 1$
- Argument: $\arg(i) = \frac{\pi}{2}$ $(90^\circ)$

## Part $2^\circ$: $z = -2i$

### Extract Real and Imaginary Parts

$$z = -2i = 0 + i \cdot (-2)$$

- Real part: $x = 0$
- Imaginary part: $y = -2$

### Corresponding Point

$$\boxed{(0, -2)}$$

### Position in Complex Plane

The point $(0, -2)$ lies on the **negative imaginary axis**.
Since $x = 0$ and $y < 0$, it is not in any quadrant but on the boundary between Quadrants III and IV.

### Additional Properties

- Modulus: $|-2i| = \sqrt{0^2 + (-2)^2} = 2$
- Argument: $\arg(-2i) = -\frac{\pi}{2}$ ($90^\circ$ or $270^\circ$)

## Part $3^\circ$: $z = -1 + i$

### Extract Real and Imaginary Parts

$$z = -1 + i = -1 + i \cdot 1$$

- Real part: $x = -1$
- Imaginary part: $y = 1$

### Corresponding Point

$$\boxed{(-1, 1)}$$

### Position in Complex Plane

The point $(-1, 1)$ lies in **Quadrant II**.
Since $x < 0$ and $y > 0$, the point has a negative real part and positive imaginary part.

### Additional Properties

- Modulus: $|-1 + i| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$
- Argument: $\arg(-1 + i) = \pi - \arctan\left(\frac{1}{1}\right) = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$ ($135^\circ$)

## Part $4^\circ$: $z = 2 - 3i$

### Extract Real and Imaginary Parts

$$z = 2 - 3i = 2 + i \cdot (-3)$$

- Real part: $x = 2$
- Imaginary part: $y = -3$

### Corresponding Point

$$\boxed{(2, -3)}$$

### Position in Complex Plane

The point $(2, -3)$ lies in **Quadrant IV**.
Since $x > 0$ and $y < 0$, the point has a positive real part and negative imaginary part.

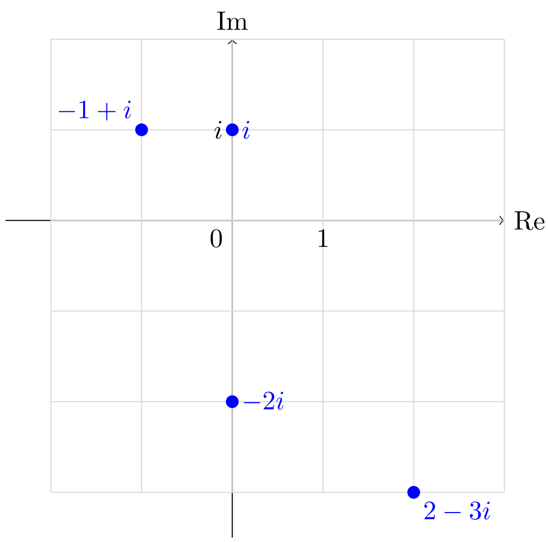
### Additional Properties

- Modulus: $|2 - 3i| = \sqrt{2^2 + (-3)^2} = \sqrt{4 + 9} = \sqrt{13}$
- Argument: $\arg(2 - 3i) = -\arctan\left(\frac{3}{2}\right) \approx -56.31$ or $\approx 303.69$

## Summary Table

| Complex Number | Point $(x, y)$ | Position | Modulus | Argument |
|---|---|---|---|---|
| $i$ | $(0, 1)$ | Positive imaginary axis | 1 | $\frac{\pi}{2}$ |
| $-2i$ | $(0, -2)$ | Negative imaginary axis | 2 | $-\frac{\pi}{2}$ |
| $-1 + i$ | $(-1, 1)$ | Quadrant II | $\sqrt{2}$ | $\frac{3\pi}{4}$ |
| $2 - 3i$ | $(2, -3)$ | Quadrant IV | $\sqrt{13}$ | $-\arctan(\frac{3}{2})$ |

## Graphical Representation




# Final Answer

$$\boxed{\begin{array}{c} 1^\circ \quad i \to (0, 1) \quad \text{(Positive imaginary axis)} \\ 2^\circ \quad -2i \to (0, -2) \quad \text{(Negative imaginary axis)} \\ 3^\circ \quad -1 + i \to (-1, 1) \quad \text{(Quadrant II)} \\ 4^\circ \quad 2 - 3i \to (2, -3) \quad \text{(Quadrant IV)} \end{array}}$$

**General Rule:** For complex number $z = x + iy$, the corresponding point is $(x, y)$.
**Position Determination:**

- If $x > 0, y > 0$: Quadrant I

- If $x < 0, y > 0$: Quadrant II
- If $x < 0, y < 0$: Quadrant III
- If $x > 0, y < 0$: Quadrant IV
- If $x = 0, y \neq 0$: On the imaginary axis
- If $x \neq 0, y = 0$: On the real axis